

Comparative Assessment: MILU vs. MMLU

Deployment context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Deployment Context

Use case: Deployment scenario: A Hindi-speaking student in India is testing his/her preparation for competitive job-related examinations, with an AI system providing feedback on their responses. The goal is to evaluate the applicability of the benchmark and the quality of the LLM's feedback.

Domain: Educational assessment

Setting: Mobile application / enterprise software

Note: The deployment hypothesis should be tested using Hindi-language sentences from the benchmark, as I am familiar with the language and can provide evaluation in a subsequent stage.

Target population: A graduate student in North India interacting in the Hindi language. The student has completed their education primarily in Hindi, with limited exposure to English.

Overall Risk Ratings

Benchmark	Risk	Avg. Score
MILU	HIGH	2.2 / 5
MMLU	HIGH	1.2 / 5

Dimension Scores Comparison

Dimension	MILU	Rating	MMLU	Rating	Δ
Input Ontology (IO)	3/5	Moderate gaps	1/5	Serious concern	+2
Input Content (IC)	2/5	Significant gaps	1/5	Serious concern	+1
Input Form (IF)	4/5	Minor gaps	1/5	Serious concern	+3
Output Ontology (OO)	1/5	Serious concern	1/5	Serious concern	0
Output Content (OC)	2/5	Significant gaps	1/5	Serious concern	+1
Output Form (OF)	1/5	Serious concern	2/5	Significant gaps	-1
Average	2.2		1.2		+1

Δ = MILU score – MMLU score. Positive = regional benchmark scores higher.

Per-Dimension Justification Comparison

Brief excerpts from each benchmark's assessment justification. Full justifications are available in the individual dimension PDFs.

Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

MILU — 3/5 (Moderate gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
MILU's 8 macro-domain / 41-subject taxonomy [Q42] aligns partially with central-exam priority subjects: Social Sciences and Law & Governance map to History/Polity, while Arts & Humanities covers cultural GK.	MMLU's 57-subject taxonomy is anchored in a US/Western academic curriculum [Q1, Q8, Q16] with subjects calibrated to US bar-exam, USMLE, and US-foreign-policy concerns [Q17, Q49].

Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

MILU — 2/5 (Significant gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
MILU's stated design priorities — India-first sourcing, regional cultural content, sourcing from real exam papers [Q11, Q12, Q24, Q25] — are well aligned with the deployment's content needs.	Individual MMLU items are culturally anchored in US/Western contexts: US Congress and constitutional history [DATASET-D1, D2, D3], US bar-exam tort scenarios [DATASET-D10, D11], Western philosophical canon [DATASET-D16, D17], US Social Security and birth-control demographics [DATASET-D20, D21], US measurement units in mathematics [DATASET-D25, D28] ...

Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

MILU — 4/5 (Minor gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
MILU is text-only with explicit exclusion of reading-comprehension passages, image-based items, and >4-option items [Q33], plus INDICLID + Unicode-based language filtering [Q34], duplicate removal [Q35], and multi-stage manual cleaning [Q31, Q32, Q36].	MMLU is entirely English-language, text-only, with a standardised English preamble for prompts [Q44] and English-only normalised input format for UnifiedQA [Q91]. Dataset metadata confirms 'languages: [en]' and monolingual content; no Devanagari script or Hindi vocabulary appears in any of the ~200 sampled items.

Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

MILU — 1/5 (Serious concern) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
This is a fundamental mismatch. MILU's output ontology is a single MCQ label — one of up to four options [Q33, Q55] — and the dataset schema confirms only a 'target' field with no rationale or explanation field (DATASET-D1, D9).	MMLU's output ontology is a four-class A/B/C/D classification [Q83, Q44] with classification accuracy as the sole performance criterion [Q34, Q23]. Dataset analysis confirms a single ClassLabel field (0–3) with no rationale field.

Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

MILU — 2/5 (Significant gaps) [medium confidence]	MMLU — 1/5 (Serious concern) [high confidence]
For the MCQ-label correctness narrowly, MILU has reasonable provenance: answers are inherited from online exam portals where 'subject experts ensure the accuracy of the answers' [Q26], with AI4Bharat volunteer manual audits [Q87] and researcher review of subject-tag clusters [Q41].	The paper is silent on annotator demographics, qualifications, IAA protocols, and label-validation procedures — Output Content concerns are explicitly NOT DOCUMENTED in the benchmark write-up. Questions were assembled by US graduate/undergraduate students [Q17] and validated against US-standard answer keys.

Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

MILU — 1/5 (Serious concern) [high confidence]	MMLU — 2/5 (Significant gaps) [high confidence]
The output form mismatch is fundamental and acknowledged by the paper itself. MILU evaluates via log-likelihood option selection [Q52, Q53, Q54, Q55] or zero-shot JSON-formatted option choice for API models [Q56, Q57].	Output form is the dimension with the strongest partial match — both MMLU and the deployment use text-based MCQ as the interaction vehicle, and the binary correct/incorrect verdict can in principle be derived from MMLU's A/B/C/D output [Q83].

Overall Summaries

MILU (risk: HIGH)

MILU is the most relevant publicly available benchmark for Indic-language knowledge evaluation and provides genuine value for a portion of this deployment's needs — particularly for measuring MCQ-level accuracy on Indian Polity, History, Geography, and Economics in Hindi. Its India-first design [Q11], real-exam sourcing [Q24, Q25], and 41-subject taxonomy [Q42] are well-suited to a central-exam preparation context in principle.

MMLU (risk: HIGH)

MMLU is a near-total mismatch for the North India Hindi-medium competitive exam deployment across five of six validity dimensions. Subject coverage is anchored in US/Western academic curriculum with zero Indian Polity, Indian History, Indian Geography, Current Affairs, or Hindi-language content (the high-priority domains identified in elicitation).